Mathematische Zeitschrift (2019) 292:1223–1231
https://doi.org/10.1007/s00209-018-2160-8

Mathematische Zeitschrift

Characterizations of toric varieties via polarized endomorphisms

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Received: 4 June 2017 / Accepted: 20 July 2018 / Published online: 13 October 2018
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Abstract
Let X be a normal projective variety and $f : X \to X$ a non-isomorphic polarized endomorphism. We give two characterizations for X to be a toric variety. First we show that if X is $\mathbb{Q}$-factorial and G-almost homogeneous for some linear algebraic group G such that f is G-equivariant, then X is a toric variety. Next we give a geometric characterization: if X is of Fano type and smooth in codimension 2 and if there is an f^{-1}-invariant reduced divisor D such that $f|_{X\setminus D}$ is quasi-étale and $K_X + D$ is $\mathbb{Q}$-Cartier, then X admits a quasi-étale cover $\widetilde{X}$ such that $\widetilde{X}$ is a toric variety and f lifts to $\widetilde{X}$. In particular, if X is further assumed to be smooth, then X is a toric variety.

Keywords Polarized endomorphism · Toric pair · Complexity

Mathematics Subject Classification 14M25 · 32H50 · 20K30 · 08A35

1 Introduction

We work over an algebraically closed field k of characteristic zero. Let X be a normal projective variety of dimension $n \geq 1$.

A surjective endomorphism $f : X \to X$ is *polarized*, if there is an ample Cartier divisor H on X such that we have $f^*H \sim qH$ (linear equivalence) for some integer $q > 1$. See [14] for some conjectures on polarised endomorphisms.

The variety X is said to be *toric* or a *toric variety* if X contains an algebraic torus $T = (k^*)^n$ as an (affine) open dense subset such that the natural multiplication action of T on itself extends to an action on the whole variety X. In this case, let $D := X\setminus T$, which is a divisor; the pair (X, D) is said to be a *toric pair*.

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We observe that a toric variety (e.g. the projective space) has lots of symmetries. The purpose of this short paper is to use symmetries to characterize toric pairs.

We first give a characterization of toric varieties via polarized endomorphisms and linear algebraic group actions. Let G be a linear algebraic group acting on X. We say X is *G-almost homogeneous* if there exists an open dense G-orbit in X.

Theorem 1.1 *Let $f : X \to X$ be a polarized endomorphism of a G-almost homogeneous normal projective variety X with G being a linear algebraic group. Assume further the following conditions.*

(i) *Let U be the open dense G-orbit in X and D the codimension-1 part of $X\backslash U$. The Weil divisor $K_X + D$ is $\mathbb{Q}$-Cartier.*
(ii) *The endomorphism f is G-equivariant in the sense: there is a surjective homomorphism $\varphi : G \to G$ such that $f \circ g = \varphi(g) \circ f$ for all g in G.*

Then (X, D) is a toric pair.

Let (X, Δ) be a log pair. The *complexity* $c = c(X, \Delta)$ of (X, Δ) is defined as

$$c := \inf \left\{ n + \dim_{\mathbb{R}} \left(\sum \mathbb{R}[S_i] \right) - \sum a_i \mid \sum a_i S_i \le \Delta, a_i \ge 0, S_i \ge 0 \right\};$$

see Definition 4.2. Brown, M^{c}Kernan, Svaldi and Zong recently gave a geometric characterization of toric varieties involving the complexity; see [4, Theorem 1.2] or Theorem 4.3. Their result is a special case of a conjecture of Shokurov, which is stated in the relative case (cf. [13]). A simple version of their result shows that *if $(X, \Delta)$is a log canonical pair such that $\Delta$is reduced, $-(K_X + \Delta)$ is nef, and $c(X, \Delta)$is non-positive, then $(X, \Delta)$is a toric pair*; see Theorem 4.3 and Remark 4.4.

A finite surjective morphism $h : Y \to Z$ between normal varieties is *quasi-étale* if h is étale outside a codimension-2 subset of Z. We show that the complexity condition of [4, Theorem 1.2] holds true in the following case.

Theorem 1.2 *Let X be a normal projective variety which is smooth in codimension 2, and $D \subset X$ a reduced divisor such that*

(i) *there is a Weil $\mathbb{Q}$-divisor Γ such that the pair (X, Γ) has only klt singularities;*
(ii) *there is a polarized endomorphism $f : X \to X$ such that D is f^{-1}-invariant and $f|_{X\backslash D}$ is quasi-étale;*
(iii) *the algebraic fundamental group $\pi_1^{\mathrm{alg}}(X_{\mathrm{reg}})$ of the smooth locus X_{reg} of X is trivial (this holds when X is smooth and rationally connected); and*
(iv) *the irregularity $q(X) := h^1(X, \mathcal{O}_X)$ is zero (this holds when X is rationally connected).*

Then the complexity $c(X, D)$ is non-positive.

Remark 1.3 Let X be an n-dimensional smooth Fano variety of Picard number one and $D \subset X$ a reduced divisor. Assume the existence of a non-isomorphic surjective endomorphism $f : X \to X$ such that D is f^{-1}-invariant and $f|_{X\backslash D}$ is étale. Hwang and Nakayama show that X is isomorphic to $\mathbb{P}^n$ and D is a simple normal crossing divisor consisting of $n + 1$ hyperplanes; see [8, Theorem 2.1]. In particular, (X, D) is a toric pair. Indeed, their argument shows that the complexity $c(X, D)$ is non-positive. Our Theorem 1.2 follows their idea and tries to generalize their result to the singular case. A key step of ours is to verify that $\hat{\Omega}^1_X(\log D)$ is free, i.e., isomorphic to $\mathcal{O}_X^{\oplus n}$; see [8, Proposition 2.3] and Theorem 5.4.

As applications of Theorem 1.2, we have the following characterizations for toric pairs. A well known conjecture asserts that projective spaces are the only smooth projective Fano varieties of Picard number one admitting an endomorphism f of degree ≥ 2 (this kind of f is automatically polarized). One may generalize it to the case of arbitrary Picard number. Below is a partial solution to it.

Corollary 1.4 *Let X be a rationally connected smooth projective variety and $D \subset X$ a reduced divisor. Suppose $f : X \to X$ is a polarized endomorphism such that D is f^{-1}-invariant and $f|_{X\setminus D}$ is étale. Then (X, D) is a toric pair.*

We say that a normal projective variety X is of *Fano type* if there is a Weil $\mathbb{Q}$-divisor Δ such that the pair (X, Δ) has only klt singularities and $-(K_X + \Delta)$ is an ample $\mathbb{Q}$-Cartier divisor. The assumption below of X being of Fano type is necessary, since a normal projective toric variety is known to be of Fano type.

Corollary 1.5 (cf. Remark 1.7) *Let $f : X \to X$ be a polarized endomorphism of a normal projective variety X of Fano type which is smooth in codimension 2. Let $D \subset X$ be an f^{-1}-invariant reduced divisor such that $f|_{X\setminus D}$ is quasi-étale and $K_X + D$ is $\mathbb{Q}$-Cartier. Then there exist a quasi-étale cover $\pi : \widetilde{X} \to X$ and a polarized endomorphism $\widetilde{f} : \widetilde{X} \to \widetilde{X}$ such that*

1. *the endomorphism f lifts to $\tilde{f}$, i.e., $\pi \circ \widetilde{f} = f \circ \pi$, and*
2. *the pair $(\widetilde{X}, \widetilde{D})$ is toric, where $\widetilde{D} = \pi^{-1}(D)$.*

The following example satisfies the conditions of both Theorems 1.1 and 1.2.

Example 1.6 Let $X := \mathbb{P}^n$ and

$$f : X \to X;\ [X_0 : \cdots : X_n] \mapsto [X_0^q : \cdots : X_n^q]$$

the power map for some $q \geq 2$. Then f is polarized with $\deg f = q^n$. Let the algebraic torus $T = (k^*)^n$ act on X naturally: $T \times X \to X$ via

$$((t_1, \ldots, t_n), [X_0 : \cdots : X_n]) \mapsto [X_0 : t_1 X_1 : \cdots : t_n X_n].$$

Let $D_i := \{X_i = 0\} \subseteq X$. Denote by $D := \sum_{i=0}^n D_i$ and $U := X\setminus D$. Let

$$\varphi : T \to T;\ (t_1, \ldots, t_n) \mapsto (t_1^q, \ldots, t_n^q)$$

which is a surjective homomorphism.

One can check easily that X is T-almost homogeneous with U the big open orbit and f is T-equivariant in the sense that $f \circ g = \varphi(g) \circ f$ for all g in T. Hence the conditions in Theorem 1.1 are all satisfied. Of course, (X, D) is a toric pair.

Note that D is f^{-1}-invariant, the restriction $f|_{X\setminus D}$ is étale, and $\pi_1(X)$ is trivial. So the conditions in Theorem 1.2 are all satisfied. Clearly, we have $c(X, D) \leq n + 1 - (n + 1) = 0$.

One may take the toric blowups of X to get more examples satisfying all conditions in Theorems 1.1 and 1.2.

Remark 1.7 In Corollary 1.5, it is not always possible to take $\pi : \widetilde{X} \to X$ to be the identity map. In other words, the pair (X, D) itself may not be toric.

Indeed, let $\widetilde{f} : \widetilde{X} = \mathbb{P}^n \to \widetilde{X}$ be the power map of degree q^n for some $q \geq 2$ as defined in Example 1.6. The symmetric group S_{n+1} in $(n + 1)$-letters acts naturally on $\mathbb{P}^n$ as (coordinates) permutations. Let $\widetilde{D}_i := \{X_i = 0\} \subset \widetilde{X}$ and $\widetilde{D} := \sum \widetilde{D}_i$. Then S_{n+1} fixes $\widetilde{D}$ (as a set). Choose a non-trivial subgroup $G \leq S_{n+1}$ such that

(i) the group G has no non-trivial pseudo-reflections (i.e. each non-trivial $g \in G$ fixes at most a codimension-2 subset) and hence the quotient map $\pi : \widetilde{X} \to X := \mathbb{P}^n/G$ is quasi-étale; and
(ii) the variety X has only terminal singularities and hence is smooth in codimension 2.

For instance, take $n > 2$ and $G = \langle (1, 2, \ldots, n+1) \rangle \cong \mathbb{Z}/(n+1)\mathbb{Z}$; see [11, Lemma 3].

Note that $g \circ \widetilde{f} = \widetilde{f} \circ g$ for any g in G. Hence $\widetilde{f}$ descends to a polarized endomorphism f on X. Let $D := \pi(\widetilde{D})$. Then we have $\pi^{-1}(D) = \widetilde{D}$ and $f^{-1}(D) = D$.

We now check that f and the pair (X, D) satisfy the assumptions of Corollary 1.5. Since both $\widetilde{f}|_{\widetilde{X}\setminus\widetilde{D}}$ and π are quasi-étale, so is $f|_{X\setminus D}$. Clearly, X is $\mathbb{Q}$-factorial. Since $K_{\widetilde{X}} = \pi^* K_X$ is anti-ample, so is K_X. Thus, X is a Fano variety and hence of Fano type.

The pair (X, D) is not toric because the number of the irreducible components of D is less than $n+1$, noting that G permutes the $\widetilde{D}_i$ non-trivially; see Remark 4.6. Of course, its quasi-étale cover $(\widetilde{X}, \widetilde{D})$ is a toric pair.

2 Preliminary results

2.1 Notation and terminology

Let X be a normal projective variety of dimension $n \geq 1$. Define:

1. $q(X) = h^1(X, \mathcal{O}_X) = \dim H^1(X, \mathcal{O}_X)$ (the irregularity); and
2. $\widetilde{q}(X) = q(\widetilde{X})$ with $\widetilde{X}$ a smooth projective model of X.

Let D_1 and D_2 be two Cartier $\mathbb{R}$-divisors on X. Denote by $D_1 \equiv D_2$ if D_1 is *numerically equivalent* to D_2, i.e., if $(D_1 - D_2) \cdot C = 0$ for every curve C on X.

Denote by X_{reg} the *smooth locus* of X. Let $U \subseteq X_{\mathrm{reg}}$ be an open dense subset. Let $\pi : Y \to X$ be a log resolution such that π is isomorphic over U and $B = Y\setminus\pi^{-1}(U)$ is a simple normal crossing (SNC) divisor. Define the *log Kodaira dimension* of U as $\bar{\kappa}(U) = \kappa(Y, K_Y + B)$ (Iitaka's D-dimension), which is independent of the choice of the compactification Y of U.

Given a reduced divisor D on X, we define the sheaf $\hat{\Omega}^1_X(\log D)$ as follows. Let $U \subseteq X$ be an open subset with $\mathrm{codim}(X - U) \geq 2$ and $D \cap U$ being a normal crossing divisor. Denote by $\Omega^1_U(\log D \cap U)$ the locally free sheaf of germs of logarithmic 1-forms on U with poles only along $U\setminus D$. Using the open immersion $j : U \hookrightarrow X$, we define

$$\hat{\Omega}^1_X(\log D) = j_* \Omega^1_U(\log D \cap U).$$

This is a reflexive coherent sheaf on X.

Throughout this paper, for a pair (X, Δ), the coefficients of Δ lie in $[0, 1]$.

The result below is frequently used and part of [12, Proposition 2.9].

Lemma 2.2 *Let $f : X \to X$ be a polarized endomorphism of a normal projective variety X. Suppose $f^*M \equiv M$ (numerical equivalence) for some Cartier $\mathbb{R}$-divisor M. Then $M \equiv 0$.*

3 Proof of Theorem 1.1

Lemma 3.1 *Let X be a normal projective variety with an algebraic torus T-action. Suppose T has a Zariski-dense open orbit U in X. Then X is a toric variety.*

Proof Let $x \in U$. Then we have $U = T/T_x$, where $T_x := \{t \in T \mid tx = x\}$. Since T is a torus, U is again a torus. For any $y \in U$ and $t \in T_x$, we have $y = t_y x$ for some $t_y \in T$. Then $ty = tt_y x = t_y tx = y$. In particular, T_x acts trivially on U and hence on X. So the natural action of $U = T/T_x$ on itself extends to X. □

Lemma 3.2 *Let X be a normal projective variety and $U \subseteq X$ an open dense subset such that U is contained in the smooth locus of X. Let D be the sum of all the prime divisors D_i contained in $X\backslash U$. Assume (X, D) is log canonical. Let $\pi : Y \to X$ be a resolution such that π is isomorphic over U and $Y\backslash\pi^{-1}(U)$ is an SNC divisor. Let D' be the strict transform of D and E the sum of π-exceptional prime divisors E_j such that $Y\backslash\pi^{-1}(U) = D' + E$. Then the log Kodaira dimension $\bar{\kappa}(U) := \kappa(Y, D' + E)$ equals $\kappa(X, K_X + D)$, and $K_Y + D' + E$ is pseudo-effective (i.e., the limit of effective $\mathbb{R}$-Cartier divisors) if and only if so is $K_X + D$.*

Proof Since (X, D) is log canonical, $K_Y + D' + E = \pi^*(K_X + D) + \sum_j b_j E_j$ with $b_j \geq 0$. Hence the lemma follows by the projection formula. □

Proof of Theorem 1.1 Write $\deg f = q^{\dim(X)}$ with $q > 1$. We may assume G is connected and acts faithfully on X. Let $x_0 \in U$. Then $U = Gx_0$ and $f(U) = f(Gx_0) = \varphi(G) f(x_0) = Gf(x_0)$. So $f(U)$ is an open dense G-orbit in X and hence $f(U) = U$. Further, we claim that $f^{-1}(U) = U$. Indeed, for any $x \notin U$, the orbit Gx has $\dim(Gx) < \dim(X)$. Since $Gf(x) = \varphi(G) f(x) = f(Gx)$, we have $\dim(Gf(x)) = \dim(Gx) < \dim(X)$. In particular, $f(x)$ is not in U. So the claim is proved. Let D be the divisorial part of $X\backslash U$. Then $f^{-1}(D) = D$.

Since U is G-transitive, $f|_U : f^{-1}(U) = U \to U$ is étale. Thus, by the logarithmic ramification divisor formula, we have $K_X + D = f^*(K_X + D)$. Hence, $K_X + D \equiv 0$ by Lemma 2.2. Since $K_X + D$ is $\mathbb{Q}$-Cartier, the pair (X, D) is log canonical by [3, Theorem 1.4]. Let $\pi : Y \to X$ be a G-equivariant log resolution of X such that $\pi|_{\pi^{-1}(U)}$ is isomorphic and $Y\backslash\pi^{-1}(U)$ is an SNC divisor (cf. [9]). Applying Lemma 3.2 and using the notation there, we have that $K_Y + D' + E$ is pseudo-effective. Note that G is linear, $\pi^{-1}(U)$ is an open dense G-orbit in Y, and G acts faithfully on Y. By [2, Theorem 1.1], G is an algebraic torus. So X is a toric variety by Lemma 3.1. Since the big torus U is an affine variety, $X\backslash U$ is of pure codimension one and hence equal to D. Thus (X, D) is a toric pair. □

4 The complexity

4.1 Some notation

Let X be an n-dimensional normal projective variety and $D = \sum_{i=1}^{\ell} D_i$ a reduced divisor on X. Denote by $\ell(D) := \ell$, the *number* of irreducible components in D; and $r(D)$ the *rank* of the vector space spanned by $D_1, \cdots, D_\ell$ in the space of Weil $\mathbb{R}$-divisors modulo algebraic equivalence.

Let (X, Δ) be a log pair. Write $\Delta = \sum_i a_i D_i$ with $a_i > 0$ and D_i distinct irreducible divisors. Denote by,

$$\langle \Delta \rangle := \llcorner \Delta \lrcorner + \ulcorner 2\Delta \urcorner - \llcorner 2\Delta \lrcorner = \sum_{i : a_i > 1/2} D_i .$$

Definition 4.2 A *decomposition* of Δ is an expression of the form,

$$\sum a_i S_i \leq \Delta,$$

where $S_i \geq 0$ are $\mathbb{Z}$-divisors and $a_i \geq 0$, $1 \leq i \leq k$. The *complexity* of this decomposition is $n + r - d$, where r is the rank of the vector space spanned by $S_1, S_2, \ldots, S_k$ in the space of Weil $\mathbb{R}$-divisors modulo algebraic equivalence and $d = \sum a_i$. The *complexity* $c = c(X, \Delta)$ of (X, Δ) is the infimum of the complexity of any decomposition of Δ.

The following theorem gives a geometric characterization of toric varieties involving the complexity by Brown, M^{c}Kernan, Svaldi and Zong.

Theorem 4.3 (cf. [4, Theorem 1.2]) *Let X be a proper variety of dimension n and let (X, Δ) be a log canonical pair such that $-(K_X+\Delta)$ is nef. If $\sum a_i S_i$ is a decomposition of complexity c less than one then there is a divisor D such that (X, D) is a toric pair, where $D \geq \langle \Delta \rangle$ and all but $\llcorner 2c \lrcorner$ components of D are elements of the set $\{S_i \mid 1 \leq i \leq k\}$.*

Remark 4.4 1. If the Δ in Theorem 4.3 is a reduced divisor with the complexity $n + r(\Delta) - \ell(\Delta) \leq 0$, then Theorem 4.3 implies that (X, Δ) itself is a toric pair.
2. Let $f : X \to X$ be a polarized endomorphism and let D be an f^{-1}-invariant reduced divisor such that $f|_{X\setminus D}$ is quasi-étale and $K_X + D$ is $\mathbb{Q}$-Cartier. Then $K_X + D \equiv 0$ by the logarithmic ramification divisor formula and Lemma 2.2. In particular, $-(K_X + D)$ is nef; further, the pair (X, D) is log canonical (cf. [3, Theorem 1.4]).

The following theorem provides us with a useful upper bound of the complexity.

Theorem 4.5 *Let X be a normal projective variety and D a reduced divisor of X. Then, in notation of 2.1 and 4.1,*

$$\ell(D) \geq h^0\left(X, \hat{\Omega}^1_X(\log D)\right) + r(D) - \tilde{q}(X).$$

In particular,

$$c(X, D) \leq n + \tilde{q}(X) - h^0\left(X, \hat{\Omega}^1_X(\log D)\right).$$

Proof Let $\pi : \widetilde{X} \to X$ be a log resolution of the pair (X, D) with E being the reduced π-exceptional divisor. Denote by $\widetilde{D}$ the largest reduced divisor contained in $\operatorname{Supp} \pi^{-1}$(non-klt locus of (X, D)). Let D' be the strict transform of D and $\widetilde{D}_i$ the irreducible component of $\tilde{D}$. Then we may write $\widetilde{D} = \sum_i \widetilde{D}_i = D' + E$.

From the exact sequence

$$0 \to \Omega^1_{\widetilde{X}} \to \Omega^1_{\widetilde{X}}\left(\log \widetilde{D}\right) \to \oplus_i \mathcal{O}_{\widetilde{D}_i} \to 0,$$

we get

$$0 \to H^0\left(\widetilde{X}, \Omega^1_{\widetilde{X}}\right) \to H^0\left(\widetilde{X}, \Omega^1_{\widetilde{X}}\left(\log \widetilde{D}\right)\right) \to \oplus_i H^0\left(\widetilde{D}_i, \mathcal{O}_{\widetilde{D}_i}\right) \to H^1\left(\widetilde{X}, \Omega^1_{\widetilde{X}}\right),$$

where the connecting homomorphism essentially sends a generator 1 of $\mathcal{O}_{\widetilde{D}_i}$ for each component $\widetilde{D}_i$ to the first Chern class $c_1(\widetilde{D}_i)$. So $\ell(\widetilde{D}) = h^0(\widetilde{X}, \Omega^1_{\widetilde{X}}(\log \widetilde{D})) + r(\widetilde{D}) - q(\widetilde{X})$.

By [6, Theorem 1.5], $\mathcal{F} := \pi_* \Omega^1_{\widetilde{X}}(\log \widetilde{D})$ is reflexive. Note that $\mathcal{F}|_U = \Omega^1_U(\log D \cap U)$, where $U \subseteq X_{\mathrm{reg}}$ such that $\operatorname{codim}(X\setminus U) \geq 2$ and $D \cap U$ is a smooth divisor. Then we have $H^0(\widetilde{X}, \Omega^1_{\widetilde{X}}(\log \widetilde{D})) = H^0(X, \mathcal{F}) = H^0(U, \mathcal{F}|_U) = H^0(X, \hat{\Omega}^1_X(\log D))$.

By the negativity lemma (cf. [1, Lemma 3.6.2]), $r(E) = \ell(E)$ and hence $r(D) \leq r(\widetilde{D}) - \ell(E)$. So the theorem is proved. □

Remark 4.6 In Theorem 4.5, if X is assumed to be $\mathbb{Q}$-factorial, then the negativity lemma implies $r(D) = r(\widetilde{D}) - \ell(E)$ at the end of the proof. In particular, we will have $\ell(D) = h^0(X, \hat{\Omega}^1_X(\log D)) + r(D) - \widetilde{q}(X)$ and $c(X, D) = n + \widetilde{q}(X) - h^0(X, \hat{\Omega}^1_X(\log D))$.

If (X, D) is assumed to be a normal projective toric pair, then it is known that $\hat{\Omega}^1_X(\log D)$ is free; see [5, 4.3, page 87]. Since X is rationally connected, $\widetilde{q}(X) = 0$. Therefore, $\ell(D) \geq n + r(D)$ (with equality holding true when X is $\mathbb{Q}$-factorial).

5 Proof of Theorem 1.2

Lemma 5.1 *Let X be a normal projective variety with finite algebraic fundamental group $\pi_1^{\mathrm{alg}}(X_{\mathrm{reg}})$. Then X admits a universal quasi-étale cover $\pi : \widetilde{X} \to X$, such that $\pi_1^{\mathrm{alg}}(\widetilde{X}_{\mathrm{reg}})$ is trivial and any surjective endomorphism f of X lifts to $\widetilde{X}$.*

Proof Since $\pi_1^{\mathrm{alg}}(X_{\mathrm{reg}})$ is finite, there is a universal quasi-étale cover $\pi : \widetilde{X} \to X$ such that $\pi_1^{\mathrm{alg}}(\widetilde{X}_{\mathrm{reg}})$ is trivial. Let W be the normalization of the fibre product of f and π and W_0 a dominant irreducible component of W. Then $W_0 \to X$ is also a quasi-étale cover. Taking the universal quasi-étale cover of W_0 which is $\widetilde{X}$, we are done. □

The same argument of [8, Proposition 2.4] gives the following.

Proposition 5.2 *Let X be a normal projective variety smooth in codimension 2 and $D \subset X$ a reduced divisor. Suppose $f : X \to X$ is a polarized endomorphism such that $f^{-1}(D) = D$ and $f|_{X\setminus D}$ is quasi-étale. Then there is a smooth open subset $U \subseteq X$ such that $D \cap U$ is a normal crossing divisor and $\mathrm{codim}(X\setminus U) \geq 3$. In particular, $\hat{\Omega}^1_X(\log D)$ is locally free over U.*

The following slightly extends [8, Propositions 2.2 and 2.3].

Proposition 5.3 *Let X be a normal projective variety which is of dimension $n \geq 2$ and smooth in codimension 2, and $D \subset X$ a reduced divisor. Suppose $f : X \to X$ is a polarized endomorphism such that $f^{-1}(D) = D$ and $f|_{X\setminus D}$ is quasi-étale. Let H be an ample divisor on X such that $f^*H \sim qH$ for some $q > 1$. Then the following hold.*

1. *The intersection numbers vanish: $c_1(\hat{\Omega}^1_X(\log D)) \cdot H^{n-1} = c_1(\hat{\Omega}^1_X(\log D))^2 \cdot H^{n-2} = c_2(\hat{\Omega}^1_X(\log D)) \cdot H^{n-2} = 0$.*
2. *The sheaf $\hat{\Omega}^1_X(\log D)$ is reflexive and H-slope semistable.*

Proof By Proposition 5.2, there is a smooth open subset $U \subseteq X$ such that $D \cap U$ is a normal crossing divisor and $\mathrm{codim}(X\setminus U) \geq 3$. Since $f|_{X\setminus D}$ is quasi-étale, $f|_{f^{-1}(U)\setminus D}$ is étale by the purity of branch loci.

There is a natural morphism $\varphi : f^*\hat{\Omega}^1_X(\log D) \to \hat{\Omega}^1_X(\log D)$ and $\varphi|_{f^{-1}(U)}$ is an isomorphism. So for $1 \leq i \leq 2$, we have

$$f^* c_i\left(\hat{\Omega}^1_X(\log D)\right) = c_i\left(f^*\hat{\Omega}^1_X(\log D)\right) = c_i\left(\hat{\Omega}^1_X(\log D)\right).$$

Then

$$\begin{aligned} & q^{n-i} c_i\left(\hat{\Omega}^1_X(\log D)\right) \cdot H^{n-i} \\ & = f^* c_i\left(\hat{\Omega}^1_X(\log D)\right) \cdot \left(f^*H\right)^{n-i} = (\deg f)\, c_i\left(\hat{\Omega}^1_X(\log D)\right) \cdot H^{n-i} \end{aligned}$$

implies $c_i(\hat{\Omega}^1_X(\log D)) \cdot H^{n-i} = 0$ for $1 \le i \le 2$, noting that $\deg f = q^n > 1$. The proof for $c_1(\hat{\Omega}^1_X(\log D))^2 \cdot H^{n-2} = 0$, is similar.

For (2), suppose the contrary that $\hat{\Omega}^1_X(\log D)$ is not H-slope semistable. Then there is a coherent subsheaf $\mathcal{F} \subset \hat{\Omega}^1_X(\log D)$ such that,

$$\mu_H(\mathcal{F}) := \frac{c_1(\mathcal{F}) \cdot H^{n-1}}{\operatorname{rank} \mathcal{F}} > 0.$$

Note that,

$$s = \sup\{\mu_H(\mathcal{F}) \mid \mathcal{F} \subset \hat{\Omega}^1_X(\log D)\} < \infty.$$

So for some $k \gg 1$, $\mu_H(g^*\mathcal{F}) = q^k \mu_H(\mathcal{F}) > s$ with $g = f^k$. Let $i : f^{-1}(U) \hookrightarrow X$ be the inclusion map and let $\mathcal{G} := i_*((g^*\mathcal{F})|_{g^{-1}(U)})$. Then $\mu_H(\mathcal{G}) = \mu_H(g^*\mathcal{F}) > s$. Note that $(g^*\mathcal{F})|_{g^{-1}(U)}$ is a subsheaf of the locally free sheaf $(g^*\hat{\Omega}^1_X(\log D))|_{g^{-1}(U)} \cong \hat{\Omega}^1_X(\log D)|_{g^{-1}(U)}$. Since $\operatorname{codim}(X \backslash g^{-1}(U)) \ge 3$ and i_* is left exact, $\mathcal{G}$ is a coherent subsheaf of the reflexive sheaf $\hat{\Omega}^1_X(\log D)$. So we get a contradiction and (2) is proved. □

Theorem 5.4 *Let X be a normal projective variety which is smooth in codimension 2, and $D \subset X$ a reduced divisor such that:*

(i) *there is a Weil $\mathbb{Q}$-divisor Γ such that the pair (X, Γ) has only klt singularities;*
(ii) *there is a polarized endomorphism $f : X \to X$ such that $f^{-1}(D) = D$ and $f|_{X \backslash D}$ is quasi-étale; and*
(iii) *the algebraic fundamental group $\pi_1^{\mathrm{alg}}(X_{\mathrm{reg}})$ of the smooth locus X_{reg} of X is trivial.*

Then $\hat{\Omega}^1_X(\log D)$ is free, i.e., isomorphic to $\mathcal{O}_X^{\oplus n}$, where $n := \dim X$.

Proof The case $n = 1$ is clear. If $n \ge 2$, apply Proposition 5.3 and [7, Theorem 1.20]. □

Proof of Theorem 1.2 If $\dim(X) = 1$, then $X \cong \mathbb{P}^1$ since $\pi_1^{\mathrm{alg}}(X_{\mathrm{reg}})$ is trivial and $\deg f > 1$. Clearly, $\ell(D) = \deg D = 2$ and $r(D) = 1$; see Remark 4.4. So $c(X, D) = 0$. Now we may assume $\dim(X) > 1$. By Theorem 5.4, $\hat{\Omega}^1_X(\log D)$ is free. Since (X, Γ) is klt, X has rational singularities. So we have $\widetilde{q}(X) = q(X) = 0$ by the assumption. Thus $c(X, D) \le \widetilde{q}(X) = 0$ (cf. Theorem 4.5). □

Proof of Corollary 1.4 By Theorem 1.2, $c(X, D) \le 0$. So (X, D) is a toric pair by Remark 4.4 and [4, Theorem 1.2] or Theorem 4.3. □

Proof of Corollary 1.5 Since X is of Fano type, there is a Weil $\mathbb{Q}$-divisor Δ such that the pair (X, Δ) is klt and $-(K_X + \Delta)$ is ample. By [7, Theorem 1.13], $\pi_1^{\mathrm{alg}}(X_{\mathrm{reg}})$ is finite. By Lemma 5.1, X admits a universal quasi-étale cover $\pi : \widetilde{X} \to X$, such that $\pi_1^{\mathrm{alg}}(\widetilde{X}_{\mathrm{reg}})$ is trivial and f lifts to $\widetilde{f}$ on $\widetilde{X}$. Note that $\widetilde{X}$, like X, is still smooth in codimension 2 by the purity of branch loci.

Let $\widetilde{D} := \pi^{-1}(D)$ and $\widetilde{\Delta} := \pi^*(\Delta)$. Then $K_{\widetilde{X}} + \widetilde{D}$ equals $\pi^*(K_X + D)$ and hence is $\mathbb{Q}$-Cartier (and numerically trivial; see Remark 4.4). Also $K_{\widetilde{X}} + \widetilde{\Delta}$ equals $\pi^*(K_X + \Delta)$ and is hence anti-ample. Note that $(\widetilde{X}, \widetilde{\Delta})$ is also klt (cf. [10, Proposition 5.20]). Therefore, $\widetilde{X}$ is also of Fano type and $q(\widetilde{X}) = 0$ by the Kawamata-Viehweg vanishing. Since $\widetilde{f}$ is the lifting of f, it is polarized and $\widetilde{f}^{-1}(\widetilde{D}) = \widetilde{D}$. Since both $\pi : \widetilde{X} \to X$ and $f|_{X \backslash D}$ are quasi-étale, so is $\widetilde{f}|_{\widetilde{X} \backslash \widetilde{D}}$.

Thus Theorem 1.2 is applicable: $c(\widetilde{X}, \widetilde{D}) \le 0$. So $(\widetilde{X}, \widetilde{D})$ is a toric pair by Remark 4.4 and [4, Theorem 1.2] or Theorem 4.3. □

Acknowledgements The second author thanks Mircea Mustata for valuable discussions and warm hospitality during his visit to Univ. of Michigan in December 2016; he is also supported by an ARF of National University of Singapore. The authors thank the referee for suggestions to improve the paper.

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